Nexora Engineering Guidelines

Coding Standards

Standard Reference

Standard ENG-STD-007, version 3.1, effective 1 April 2025. Classification: Internal - Engineering. Owner: Engineering Manager.

Language Conventions

Python services follow PEP 8 and must use type hints on every public function.

TypeScript code is compiled in strict mode. Go services are formatted with gofmt.

Functions should stay under 50 lines, and each module should have a single responsibility.

Secrets and Documentation

Secrets are never committed to source control. They are stored in the managed secrets vault, and pre-commit secret scanning is mandatory on every repository.

Every public API is documented with an OpenAPI specification.

Git Workflow and Code Review

Git Workflow

Nexora uses trunk-based development on the main branch. Branches are named feature/<ticket-id> or fix/<ticket-id> and must be merged within 3 working days.

Pull requests are squash merged. Signed commits are required for production repositories.

Code Review Policy

Every pull request to a production service requires at least two approving reviews. Internal tools require one approving review.

Authors cannot approve their own pull requests. Reviewers respond within 1 business day.

Changes to authentication, authorization or cryptography code require an additional reviewer from the Security department.

Pull requests should change fewer than 400 lines.

Testing Requirements

Coverage

New code must reach at least 80 percent line coverage. Financial calculation code in Nexora Ledger must reach 95 percent line coverage.

Test Types

Every service has unit tests and integration tests that run against a real PostgreSQL database. End-to-end smoke tests run before every release.

Continuous Integration

Continuous integration must pass before merge. A flaky test is quarantined within 2 days and fixed within 10 working days.

Changes that can affect latency require a performance test. The p95 latency budget for the Nexora Ledger API is 300 milliseconds.

Release Process and Deployment Guidelines

Release Train

Production releases ship weekly every Wednesday at 11:00 IST. The change freeze starts on Tuesday at 18:00 IST.

There are no production deployments on Fridays or public holidays, except hotfixes approved by an engineering manager.

Release Requirements

A release requires a green continuous integration run, release notes, a documented rollback plan and sign-off from the QA lead and the engineering manager.

Deployment Guidelines

Services use blue-green deployments. A canary release receives 10 percent of traffic for 30 minutes before full rollout.

Deployments roll back automatically if the error rate exceeds 2 percent.

Database migrations must stay backward compatible for one release.

Incident Escalation and Quality Standards

Production Incident Priorities

P1 (full outage or data corruption): the on-call engineer responds within 15 minutes, the engineering manager is paged within 30 minutes and status updates are posted every 30 minutes.

P2 (major degradation): response within 30 minutes. P3 (minor issue): next business day. P4 (cosmetic issue): added to the backlog.

Postmortems and Handover

A blameless postmortem is completed for every P1 and P2 incident within 5 business days.

Incidents with a suspected security cause are handed over to the Security department immediately.

Quality Standards

The availability service level objective for Nexora Ledger and Nexora Flow is 99.9 percent per month. If the error budget is exhausted, feature releases are frozen until reliability work restores it.